Hanyi Seok

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Research Interests: Carbon capture and conversion; electrochemical separations and resource recovery; reaction–transport modeling and measurement design

EDUCATION

Dual-degree program between Stony Brook University and the University of Seoul

Stony Brook University *Expected May 2027*
B.E. in Engineering Science, specialization in Environmental Engineering
Second Major in Applied Mathematics and Statistics
GPA: 3.97/4.00

University of Seoul *Expected May 2027*
B.E. in Environmental Engineering
GPA: 4.10/4.50

RESEARCH EXPERIENCE

Stony Brook University — Seo Research Group **Stony Brook, NY**
Undergraduate Researcher | Advisor: Prof. Hyowon Seo *May 2025 – Present*
Hollow-Fiber Membrane Contactor Design and Modeling [Project PDF]

- Designed and fabricated a hollow-fiber membrane contactor prototype for alkaline CO_2 absorption using Fusion 360 and polypropylene 3D printing; adjusted printing parameters and assessed sealing, tubing, and fiber-potting options.
- Developed a radial–axial finite-volume model of CO_2 absorption and pH evolution, coupling species-specific diffusion and electromigration, carbonate equilibrium, gas transport, and tank balances. Tracked dissolved inorganic carbon and alkalinity; checked carbon conservation and sensitivity to mesh and time-step size.
- Independently varied gas flow and an apparent mass-transfer coefficient in a separate simplified HFMC model; showed that operating points with similar times to a target pH respond differently to increases in flow or transfer capacity.

NTU-Based Analysis of CO_2 Loading Time [Project PDF]

- Derived an expression for time to a target pH using the number of transfer units (NTU), gas-phase CO_2 depletion, carbonate equilibrium, and the recirculating-tank balance.
- Compared the expression with simulations resolving within-pass CO_2 accumulation at matched transfer coefficients; linked larger discrepancies at low liquid flow and near equilibrium to reduced local driving force.
- Derived a first-order correction for within-pass CO_2 accumulation, reducing absolute loading-time discrepancies against the numerical reference across all 100 tested condition–target combinations; relative time error fell from 11.8% to 2.0% in one low-flow case.

Northeastern University — Barecka Lab [Project PDF] **Remote**
Undergraduate Researcher | Advisor: Prof. Magda H. Barecka *Jun 2026 – Aug 2026*

- Developed carbon and isotope balances to model ^{13}C enrichment in unreacted CO_2 during electrolysis, coupling absorption, reaction, membrane transport, and liquid and product carbon inventories. Represented isotope exchange even when opposing fluxes gave zero net carbon transfer.
- Fitted a reduced absorption model to archived pH data from 27 zero-current bubbling conditions (median fitting RMSE: 0.180 pH units); separately showed that baseline selection changed inferred responses in archived outlet-isotope data.
- Analyzed published anode-gas data: assuming 100% or 95% of the current contributed to oxygen production yielded similar fits but different estimates of an effective bicarbonate–carbonate transport factor. Proposed independent oxygen, charge, and liquid-carbon measurements to distinguish these interpretations.
- Implemented eight Python/Jupyter parameter estimators for gas–liquid exchange, membrane transport, and product carbon sources. Synthetic tests covered recovery, held-out predictions, and uncertainty: a prescribed source fraction was recovered under fixed isotope assumptions, while relative membrane current-carrier weights remained poorly separated.

Korea Institute of Energy Technology — Wooyul Kim Lab [Project PDF] **Naju, South Korea**
Undergraduate Researcher | Advisor: Prof. Wooyul Kim *Jul 2026 – Aug 2026*

- Prepared metal-modified CeO_2 nanorod supports by alkaline hydrothermal synthesis and deposited nominally 2 wt% Ru via $NaBH_4$ reduction for a team study of photothermal CO_2 hydrogenation.
- Screened seven CeO_2 support modifiers using photothermal reaction tests and gas chromatography (GC); Ru/Zr–CeO_2 gave the highest CH_4 response in the initial batch, with approximately 20% higher peak area than Ru/CeO_2 after 1 h.
- Expanded Zr-content screening based on GC results; the highest CH_4 response in the expanded batch occurred at the upper tested limit of nominally 30 mol%.

Rutgers University — Dismukes Research Group [Project PDF] **Remote**
Undergraduate Researcher | Advisor: Prof. G. Charles Dismukes *Jun 2026 – Aug 2026*

- Built a reproducible Python workflow for 116 microscopy images from 20 bismuth-deposited Toray electrodes and five pristine specimens at four magnifications; combined repeated views within each specimen and magnification before statistical testing.
- Compared deposited and pristine specimens using permutation tests, bootstrap confidence intervals, and multiple-comparison correction. Found higher mean brightness at 260× (adjusted $p = 0.0024$) and a smaller gap between median and 95th-percentile brightness at all four magnifications.
- Built an exploratory watershed pipeline to complement whole-image statistics with spatial measurements of locally bright regions in 260× carbon-fiber images; reported candidate counts, sizes in pixels, and quality-control flags.

University of Seoul — Water Environment and Engineering Lab [Project PDF] **Seoul, South Korea**
Undergraduate Researcher | Advisor: Prof. Ihnsup Han *May 2024 – Jan 2025*

- Designed and operated a laboratory-scale batch reactor combining electrochemical treatment and granular activated-carbon adsorption for synthetic monoethanolamine (MEA) wastewater relevant to CO_2 capture.
- Compared electrode materials (stainless steel, Ti, Al), voltage, spacing, carbon dosage, initial MEA concentration, and temperature; monitored MEA removal and nitrogen species to examine potential removal pathways.
- Selected Al for follow-up testing based on electrode cost, literature evidence, and initial single-run screening: 67.3% MEA removal after 120 min versus 69.6% for stainless steel. Used subsequent ANOVA/Tukey comparisons to recommend operating conditions within the tested range.

University of Seoul — Environmental Systems Toxicology Lab [Project PDF] **Seoul, South Korea**
Undergraduate Researcher | Advisor: Prof. Jinhee Choi *Jun 2021 – Feb 2022*

- Linked in vitro assay activity from ToxCast with rat developmental and reproductive toxicity records from ToxRef; integrated 449 chemicals and 949 assays by chemical identifier, distinguishing inactive results from missing observations.
- Analyzed correlations in R between binary assay activity and log-transformed dose values for each toxicity endpoint; shortlisted eight associations ($|r| \geq 0.5$; $n \geq 15$) and selected two G protein-coupled receptor assays for biological review.

SELECTED PROJECTS

Dissolved Oxygen Prediction for Anyang Stream [Project PDF]
Smart Water Environment Management *Sep 2024 – Dec 2024*

- Developed a dissolved oxygen prediction workflow using 17,965 water-quality records and five predictors; selected MICE–KNN ensemble imputation after comparing four approaches under 10–30% simulated missingness.
- Compared six predictive models and tuned an LSTM network, achieving $R^2 = 0.915$ on the test set and 0.966 on subsequent November 2024 observations not used for training.
- Ranked first among all course projects and presented the project at the course showcase.

Sustainable Aviation Fuel Supply Roadmap for Korea [Project PDF]
Unit Design of Processes in Environmental Engineering *May 2024 – Jul 2024*

- Led an eight-member team in comparing four production pathways for a Korean sustainable aviation fuel (SAF) supply roadmap.
- Estimated annual domestic production potential of 734.6 million L of bio-based SAF and 268.0 million L of synthetic SAF using feedstock energy content, conversion efficiencies, and carbon-content calculations.
- Projected a 2050 synthetic-SAF shortfall of 254.3 million L using 2009–2021 consumption trends and European blending benchmarks; examined direct air capture, hydrogen supply, and fuel-price scenarios. The project ranked first among eight teams.

University of Seoul Campus GHG Inventory and 2040 Net-Zero Roadmap [Project PDF]
Unit Design of Processes in Environmental Engineering *Mar 2024 – May 2024*

- Led an eight-member team in constructing a 2023 campus GHG inventory using institutional records and Korean/IPCC emission factors; estimated net emissions of approximately 14,900 tCO_2e after forest uptake, with purchased electricity the largest source.
- Analyzed 46 commuting-survey responses using mode-specific emission factors, round-trip distances, and annual commuting frequency; scaled group-level estimates to student, faculty, and staff populations.
- Proposed a phased roadmap addressing electricity supply and demand through renewable generation and microgrids, alongside waste-reduction incentives and financing options; targeted a 50% reduction by 2030 and net zero by 2040. The project ranked first among eight teams.

PRESENTATIONS

H. Seok and D. Jeong. *Process Optimization for Treatment of Amine-Rich Wastewater Generated from Wet Carbon Capture.* Undergraduate Thesis Presentation, Urban Sciences College, University of Seoul, South Korea, 2025.

H. Seok, S. Jang, and Y. Baek. SELFIES ALPHA: An Index Based Notation Model with Optical Isomer Symbols for Improved Recognition Capacity within Toxicity Prediction. Poster, Environmental Health Policy Idea Competition, South Korea, 2023.

H. Seok, M. Jo, and S. Ahn. *Identification of ToxCast Assays for Reproductive Toxicity Screening by Correlation Analysis of ToxCast and ToxRef Data.* Research Workshop, Graduate School of Chemical Safety Management, University of Seoul, South Korea, 2022.

HONORS & AWARDS

Tau Beta Pi Engineering Honor Society — *Fall 2025*

Dean's List, Stony Brook University — *Spring 2025; Fall 2025; Spring 2026*

Korea–U.S. Advanced Technology Youth Exchange Program Scholarship (KIAT) — *Spring 2025 – Fall 2025*

Presidential Scholarship, University of Seoul — *Spring 2021 – Fall 2024*

Outstanding Student Scholarship, Digital Water Industry Innovation Talent Development Program — *Spring 2024*

Gold Prize, 2023 Environmental Health Policy Idea Competition (1st of 15 teams) — *Nov 2023*

Graduate School Dean's Award, University of Seoul (2nd of 20 teams) — *Feb 2022*

TECHNICAL SKILLS

Programming: Python, R

Modeling & Simulation: Reaction–transport and compartmental modeling; finite-volume methods; parameter estimation and identifiability; sensitivity and uncertainty analysis

Data Analysis & Statistics: Statistical inference, LSTM, missing-data imputation, microscopy image analysis

Experimental Methods: Electrochemical wastewater treatment, catalyst synthesis and Ru deposition, photothermal reaction testing, gas chromatography, cyclic voltammetry

CAD & Fabrication: Fusion 360, SolidWorks, 3D printing

ADDITIONAL EXPERIENCE

Republic of Korea Army — **Yangpyeong, South Korea**

Logistics Clerk | Squad Leader — *Apr 2022 – Oct 2023*

- Developed Python scripts to clean, cross-check, and summarize administrative records for weekly reporting; automated printing of a logistics list containing more than 1,000 items using VBS.
- Led a 12-soldier squad, coordinating training schedules, safety procedures, personnel documentation, and performance evaluations.